



# Bootstrap framework

Môn học:

Web Design[ *Buổi 6* ]

Bộ môn:

Khoa học máy tính



# Introduce to **Bootstrap**



# History

- Bootstrap was developed by **Mark Otto** and **Jacob Thornton** at Twitter, and released as an open source product in August 2011 on GitHub.
- **In June 2014 Bootstrap was the No.1 project on GitHub!**

Mark Otto



Jacob Thornton



# What is **Bootstrap**?

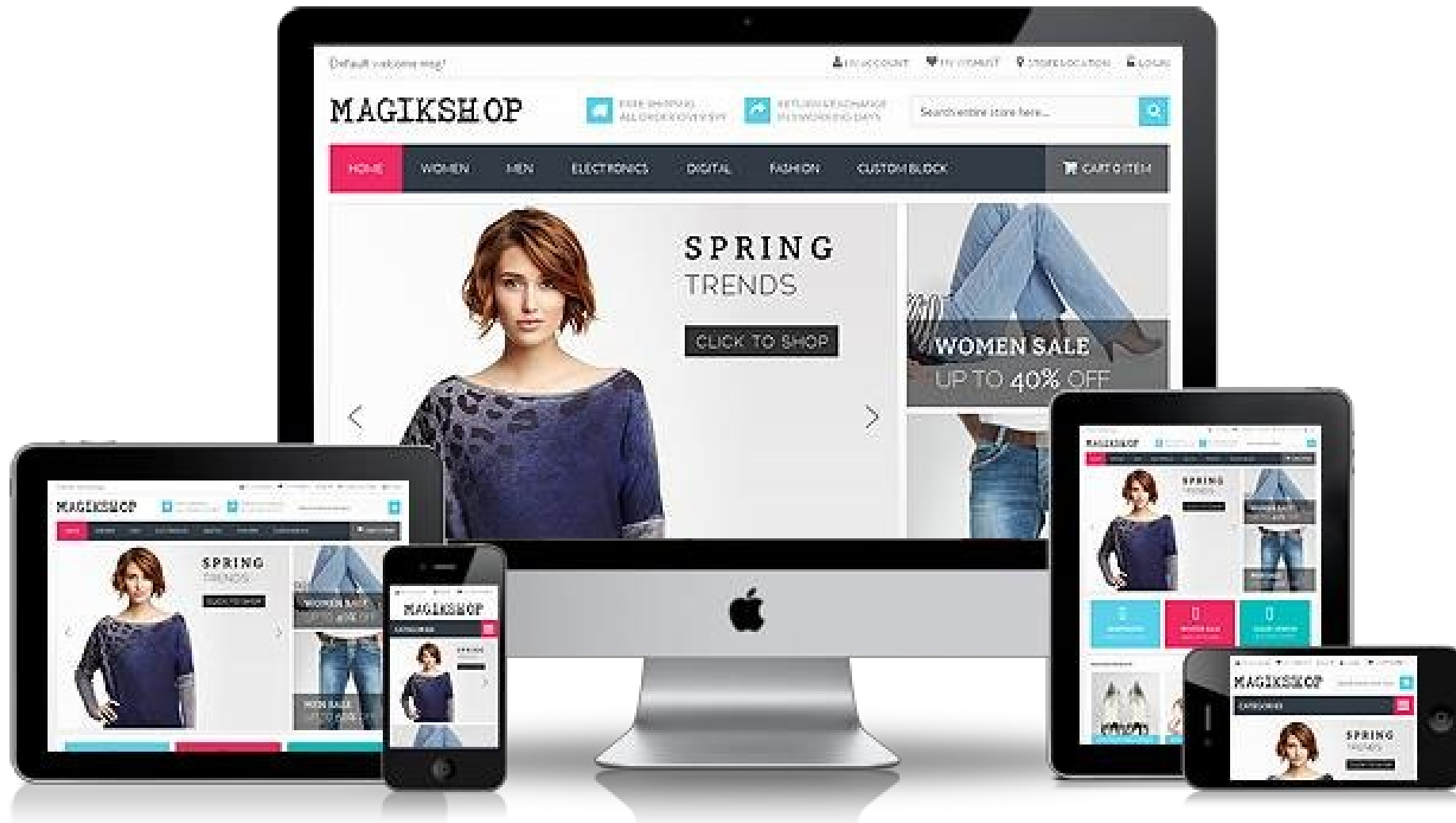
- Bootstrap is a free **front-end framework** for faster and easier web development
- Bootstrap **includes HTML and CSS based** design templates for typography, forms, buttons, tables, navigation, modals, image carousels and many other, as well as optional JavaScript plugins
- Bootstrap also gives you the ability to easily create **responsive designs**

# What is **Responsive** Web Design?

Responsive web design is about creating web sites which automatically adjust themselves to look good on all devices, from small phones to large desktops.



# Responsive Web Design



# Advantages of **Bootstrap**

- ✓ **Easy to use:** Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
- ✓ **Responsive features:** Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
- ✓ **Mobile-first approach:** In Bootstrap 3, mobile-first styles are part of the core framework
- ✓ **Browser compatibility:** Bootstrap is compatible with all modern browsers (*Chrome, Firefox, Internet Explorer, Edge, Safari, and Opera*)

# Where to Get **Bootstrap**?

There are two ways to start using Bootstrap on your own web site.

- ❖ Download Bootstrap from **getbootstrap.com**
- ❖ Include Bootstrap from a CDN (*Content Delivery Network*)

# Bootstrap **CDN**

If you don't want to download and host Bootstrap yourself, you can include it from a CDN (*Content Delivery Network*). MaxCDN provides CDN support for Bootstrap's CSS and JavaScript. You must also include jQuery:

MaxCDN:

```
<!-- Latest compiled and minified CSS -->
<link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">

<!-- jQuery library -->
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.3.1/jquery.min.js"></script>

<!-- Latest compiled JavaScript -->
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>
```

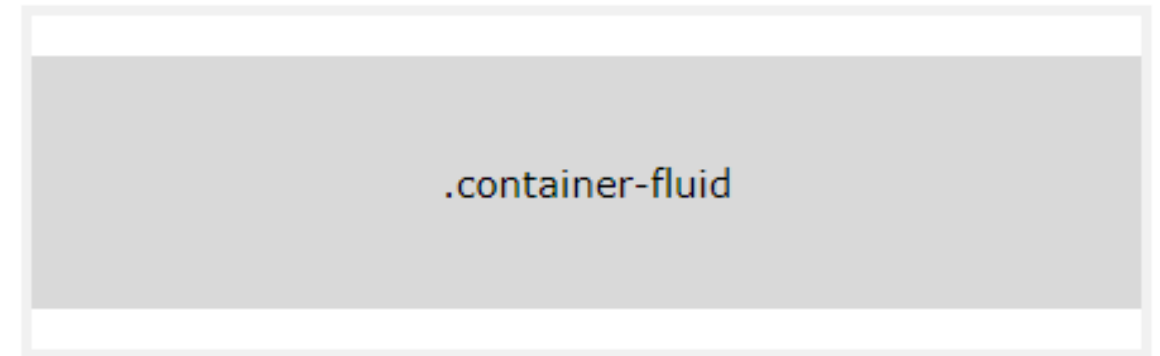
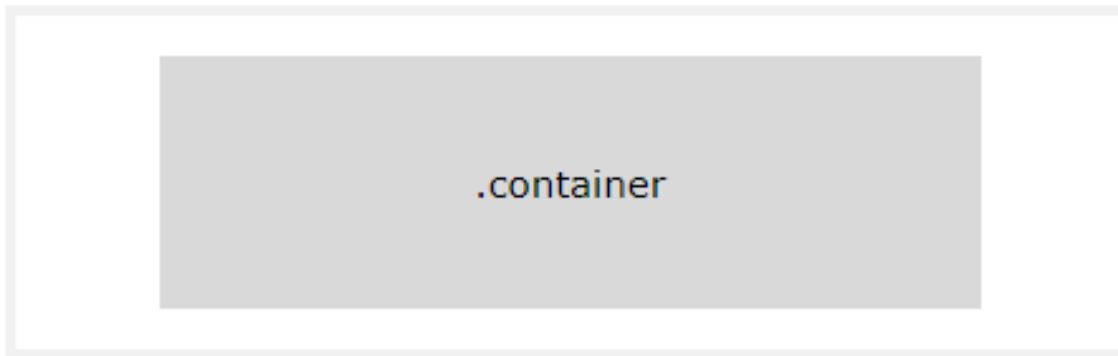
# Advantage of using the **Bootstrap CDN**

- Many users already have downloaded Bootstrap from MaxCDN when visiting another site. As a result, it will be loaded from cache when they visit your site, which leads to faster loading time. Also, most CDN's will make sure that once a user requests a file from it, it will be served from the server closest to them, which also leads to faster loading time.

# Create First Web Page With Bootstrap

1. Add the HTML5 doctype
2. Bootstrap is mobile-first
3. Containers

1. The `.container` class provides a responsive **fixed width container**
2. The `.container-fluid` class provides a **full width container**, spanning the entire width of the viewport

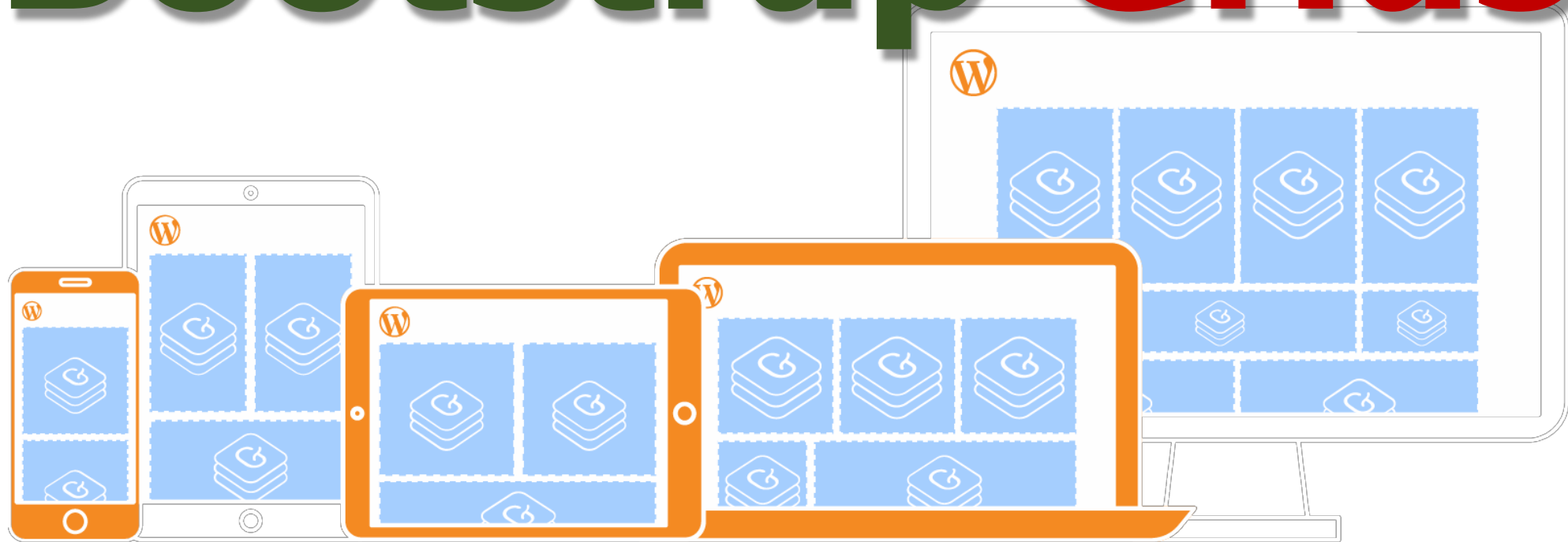


# Simple page using **Bootstrap**

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Bootstrap Example</title>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
    <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.3.1/jquery.min.js"></script>
    <script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>
  </head>
  <body>
    <div class="container">
      <h1>My First Bootstrap Page</h1>
      <p>This is some text.</p>
    </div>
  </body>
</html>
```

[https://www.w3schools.com/bootstrap/bootstrap\\_get\\_started.asp](https://www.w3schools.com/bootstrap/bootstrap_get_started.asp)

# Bootstrap Grids



# Bootstrap Grid System

Bootstrap's grid system allows up to 12 columns across the page. If you do not want to use all 12 columns individually, you can group the columns together to create wider columns

Bootstrap's grid system is responsive, and the columns will re-arrange automatically depending on the screen size.

|         |        |        |        |        |        |        |        |        |        |        |        |
|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| span 1  | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 | span 1 |
| span 4  |        |        |        | span 4 |        |        |        | span 4 |        |        |        |
| span 4  |        |        |        | span 8 |        |        |        |        |        |        |        |
| span 6  |        |        |        |        |        | span 6 |        |        |        |        |        |
| span 12 |        |        |        |        |        |        |        |        |        |        |        |

# Grid Classes

The Bootstrap grid system has four classes:

- ❖ **xs** (*for phones - screens less than 768px wide*)
- ❖ **sm** (*for tablets - screens equal to or greater than 768px wide*)
- ❖ **md** (*for small laptops - screens equal to or greater than 992px wide*)
- ❖ **lg** (*for laptops and desktops - screens equal to or greater than 1200px wide*)

The classes above can be combined to create more dynamic and flexible layouts.

# Basic Structure of a Bootstrap Grid

```
<div class="row">
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
</div>
<div class="row">
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
</div>
<div class="row">
  ...
</div>
```

First; create a row (**<div class="row">**). Then, add the desired number of columns (*tags with appropriate **.col-\*-\*** classes*). Note that numbers in **.col-\*-\*** should always add up to 12 for each row

[https://www.w3schools.com/bootstrap/bootstrap\\_grid\\_basic.asp](https://www.w3schools.com/bootstrap/bootstrap_grid_basic.asp)

# Three Equal Columns

The following example shows how to get a three equal-width columns starting at tablets and scaling to large desktops. On mobile phones or screens that are less than 768px wide, the columns will automatically stack:

```
<div class="row">
  <div class="col-sm-4">.col-sm-4</div>
  <div class="col-sm-4">.col-sm-4</div>
  <div class="col-sm-4">.col-sm-4</div>
</div>
```

.col-sm-4

.col-sm-4

.col-sm-4

# Two Unequal Columns

The following example shows how to get two various-width columns starting at tablets and scaling to large desktops:

```
<div class="row">  
  <div class="col-sm-4">.col-sm-4</div>  
  <div class="col-sm-8">.col-sm-8</div>  
</div>
```

|           |           |
|-----------|-----------|
| .col-sm-4 | .col-sm-8 |
|-----------|-----------|

# Bootstrap Glyphicons



## 290+ Glyphicons and Zocial Icons

Readable includes 290+ resizable icons from Glyphicons and Zocial icons.

Bootstrap provides 260 glyphs from the [Glyphicons](#) Halflings set. Glyphicons can be used in text, buttons, toolbars, navigation, forms, etc.

Here are some examples of glyphs:

Envelope glyph: 

Print glyph: 

Search glyph: 

Download glyph: 

[https://www.w3schools.com/bootstrap/bootstrap\\_glyphicons.asp](https://www.w3schools.com/bootstrap/bootstrap_glyphicons.asp)

# Glyphicons



# How to use Glyphicons

## Glyphicon Examples

Envelope icon: ✉

Envelope icon as a link: ✉

Search icon: 🔍

Search icon on a button: 🔍 Search

Search icon on a styled button: 🔍 Search

Print icon: 🖨

Print icon on a styled link button: 🖨 Print

</p>

# HTML Elements in Bootstrap



# Bootstrap **Text/Typography**

## ■ Bootstrap's Default Settings

- Bootstrap's global **default font-size** is **14px**, with a **line-height** of **1.428**.
- This is applied to the **<body>** element and all paragraphs (**<p>**).
- In addition, all **<p>** elements have a bottom margin that equals half their computed line-height (*10px by default*).

## ■ Bootstrap will style the HTML headings (**<h1>** to **<h6>**) in the following way:

h1 Bootstrap heading (36px)

h2 Bootstrap heading (30px)

h3 Bootstrap heading (24px)

h4 Bootstrap heading (18px)

h5 Bootstrap heading (14px)

h6 Bootstrap heading (12px)

[https://www.w3schools.com/bootstrap/bootstrap\\_typography.asp](https://www.w3schools.com/bootstrap/bootstrap_typography.asp)

# Bootstrap **Typography** Classes

| Class                         | Description                                                                                                                                                                                                                                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>.text-capitalize</code> | Indicates capitalized text                                                                                                                                                                                                                                                                         |
| <code>.initialism</code>      | Displays the text inside an <code>&lt;abbr&gt;</code> element in a slightly smaller font size                                                                                                                                                                                                      |
| <code>.list-unstyled</code>   | Removes the default list-style and left margin on list items (works on both <code>&lt;ul&gt;</code> and <code>&lt;ol&gt;</code> ). This class only applies to immediate children list items (to remove the default list-style from any nested lists, apply this class to any nested lists as well) |
| <code>.list-inline</code>     | Places all list items on a single line                                                                                                                                                                                                                                                             |
| <code>.dl-horizontal</code>   | Lines up the terms ( <code>&lt;dt&gt;</code> ) and descriptions ( <code>&lt;dd&gt;</code> ) in <code>&lt;dl&gt;</code> elements side-by-side. Starts off like default <code>&lt;dl&gt;</code> s, but when the browser window expands, it will line up side-by-side                                 |
| <code>.pre-scrollable</code>  | Makes a <code>&lt;pre&gt;</code> element scrollable                                                                                                                                                                                                                                                |

[https://www.w3schools.com/bootstrap/bootstrap\\_typography.asp](https://www.w3schools.com/bootstrap/bootstrap_typography.asp)

# Bootstrap **Images**

## ■ Class for image shapes

Rounded Corners:



**.img-rounded**

Circle:



**.img-circle**

Thumbnail:



**.img-thumbnail**

Example

```

```

# Bootstrap **List Groups**

The most basic list group is an unordered list with list items:

|             |
|-------------|
| First item  |
| Second item |
| Third item  |

To create a basic list group, use an **<ul>** element with class **.list-group**, and **<li>** elements with class **.list-group-item**:

```
<ul class="list-group">  
  <li class="list-group-item">First item</li>  
  <li class="list-group-item">Second item</li>  
  <li class="list-group-item">Third item</li>  
</ul>
```

[https://www.w3schools.com/bootstrap/bootstrap\\_list\\_groups.asp](https://www.w3schools.com/bootstrap/bootstrap_list_groups.asp)

# Bootstrap **Dropdowns**

- A dropdown menu is a toggleable menu that allows the user to choose one value from a predefined list:
- The **.dropdown** class indicates a dropdown menu. To open the dropdown menu, use a button or a link with a class of **.dropdown-toggle** and the **data-toggle="dropdown"** attribute. The **.caret** class creates a caret arrow icon (), which indicates that the button is a dropdown. Add the **.dropdown-menu** class to a **<ul>** element to actually build the dropdown menu.

[https://www.w3schools.com/bootstrap/bootstrap\\_dropdowns.asp](https://www.w3schools.com/bootstrap/bootstrap_dropdowns.asp)



# Bootstrap **Dropdowns** example

```
<div class="dropdown">
  <button class="btn btn-primary dropdown-toggle" type="button" data-toggle="dropdown">
    Dropdown Example <span class="caret"></span></button>
  <ul class="dropdown-menu">
    <li><a href="#">HTML</a></li>
    <li><a href="#">CSS</a></li>
    <li><a href="#">JavaScript</a></li>
    <li class="divider"></li>
    <li><a href="#">About Us</a></li>
  </ul>
</div>
```

Dropdown Example ▾

HTML

CSS

JavaScript

About Us

# Bootstrap **Tables**

A basic Bootstrap table has a light padding and only horizontal dividers.  
The **.table** class adds basic styling to a table:

## Example

| Firstname | Lastname | Email            |
|-----------|----------|------------------|
| John      | Doe      | john@example.com |
| Mary      | Moe      | mary@example.com |
| July      | Dooley   | july@example.com |

**.table-striped**      **.table-bordered**      **.table-hover**

[https://www.w3schools.com/bootstrap/bootstrap\\_tables.asp](https://www.w3schools.com/bootstrap/bootstrap_tables.asp)

# Bootstrap **Tables**

Contextual classes can be used to color table rows (`<tr>`) or table cells (`<td>`):

| Class                 | Description                                            |
|-----------------------|--------------------------------------------------------|
| <code>.active</code>  | Applies the hover color to the table row or table cell |
| <code>.success</code> | Indicates a successful or positive action              |
| <code>.info</code>    | Indicates a neutral informative change or action       |
| <code>.warning</code> | Indicates a warning that might need attention          |
| <code>.danger</code>  | Indicates a dangerous or potentially negative action   |

[https://www.w3schools.com/bootstrap/bootstrap\\_tables.asp](https://www.w3schools.com/bootstrap/bootstrap_tables.asp)

# Bootstrap **Responsive Tables**

The **.table-responsive** class creates a responsive table. The table will then scroll horizontally on small devices (*under 768px*). When viewing on anything larger than 768px wide, there is no difference:

## Example

```
<div class="table-responsive">
  <table class="table">
    ...
  </table>
</div>
```

[https://www.w3schools.com/bootstrap/bootstrap\\_tables.asp](https://www.w3schools.com/bootstrap/bootstrap_tables.asp)

# Bootstrap **Forms**

Form controls automatically receive some global styling with Bootstrap:

All textual **<input>**, **<textarea>**, and **<select>** elements with class **.form-control** have a width of 100%.

Bootstrap provides three types of form layouts:

- ✓ Vertical form (*this is default*)
  - ✓ Horizontal form
  - ✓ Inline form

Standard rules for all three form layouts:

- ✓ Wrap labels and form controls in **<div class="form-group">**  
(*needed for optimum spacing*)
- ✓ Add class **.form-control** to all textual **<input>**, **<textarea>**, and **<select>** elements

[https://www.w3schools.com/bootstrap/bootstrap\\_forms.asp](https://www.w3schools.com/bootstrap/bootstrap_forms.asp)

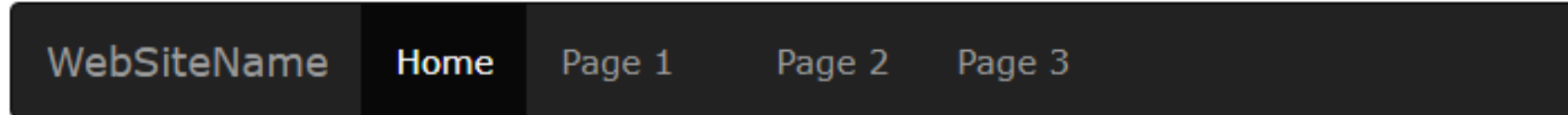
# Bootstrap **Forms**

```
<form action="/action_page.php">
  <div class="form-group">
    <label for="email">Email address:</label>
    <input type="email" class="form-control" id="email">
  </div>
  <div class="form-group">
    <label for="pwd">Password:</label>
    <input type="password" class="form-control" id="pwd">
  </div>
  <div class="checkbox">
    <label><input type="checkbox"> Remember me</label>
  </div>
  <button type="submit" class="btn btn-default">Submit</button>
</form>
```

  
  
☐ Remember me

# Bootstrap Navigation Bar

Anavigation bar is a navigation header that is placed at the top of the page:



With Bootstrap, a navigation bar can extend or collapse, depending on the screen size.

A standard navigation bar is created with `<nav class="navbar navbar-default">`.

The following example shows how to add a navigation bar to the top of the page:

```
<nav class="navbar navbar-default">
  <div class="container-fluid">
    <div class="navbar-header">
      <a class="navbar-brand" href="#">WebSiteName</a>
    </div>
    <ul class="nav navbar-nav">
      <li class="active"><a href="#">Home</a></li>
      <li><a href="#">Page 1</a></li>
      <li><a href="#">Page 2</a></li>
      <li><a href="#">Page 3</a></li>
    </ul>
  </div>
</nav>
...
```

[https://www.w3schools.com/bootstrap/bootstrap\\_navbar.asp](https://www.w3schools.com/bootstrap/bootstrap_navbar.asp)

# Bootstrap **Carousel Plugin**

The Carousel plugin is a component for cycling through elements, like a carousel (slideshow).

- ❖ The outermost <div>
- ❖ The "Indicators" part
- ❖ The "Wrapper for slides" part
- ❖ The "Left and right controls" part

**CSS Class:**

**.carousel .slide .carousel-indicators**  
**.carousel-inner .active .item**



[https://www.w3schools.com/bootstrap/bootstrap\\_carousel.asp](https://www.w3schools.com/bootstrap/bootstrap_carousel.asp)

# Bootstrap Carousel Plugin

```
<!-- Left and right controls -->
<a class="left carousel-control" href="#myCarousel" data-slide="prev">
  <span class="glyphicon glyphicon-chevron-left"></span>
  <span class="sr-only">Previous</span>
</a>
<a class="right carousel-control" href="#myCarousel" data-slide="next">
  <span class="glyphicon glyphicon-chevron-right"></span>
  <span class="sr-only">Next</span>
</a>
</div>

<div class="carousel-caption">
  <h3>New York</h3>
  <p>We love the Big Apple!</p>
</div>
</div>
</div>
```



[https://www.w3schools.com/bootstrap/bootstrap\\_carousel.asp](https://www.w3schools.com/bootstrap/bootstrap_carousel.asp)

# Bootstrap **Buttons** Plugin

Bootstrap provides different styles of buttons:



To achieve the button styles above, Bootstrap has the following classes:

- .btn**
- .btn-default**
- .btn-primary**
- .btn-success**
- .btn-info**
- .btn-warning**
- .btn-danger**
- .btn-link**

```
<button type="button" class="btn">Basic</button>
<button type="button" class="btn btn-default">Default</button>
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-link">Link</button>
```

[https://www.w3schools.com/bootstrap/bootstrap\\_buttons.asp](https://www.w3schools.com/bootstrap/bootstrap_buttons.asp)

# Bootstrap Templates



Blog:

3 col

9 col

Footer

[Try it Yourself »](#)

## Portfolio:

Navbar

Jumbotron

3 col

3 col

3 col

3 col

3 col

3 col

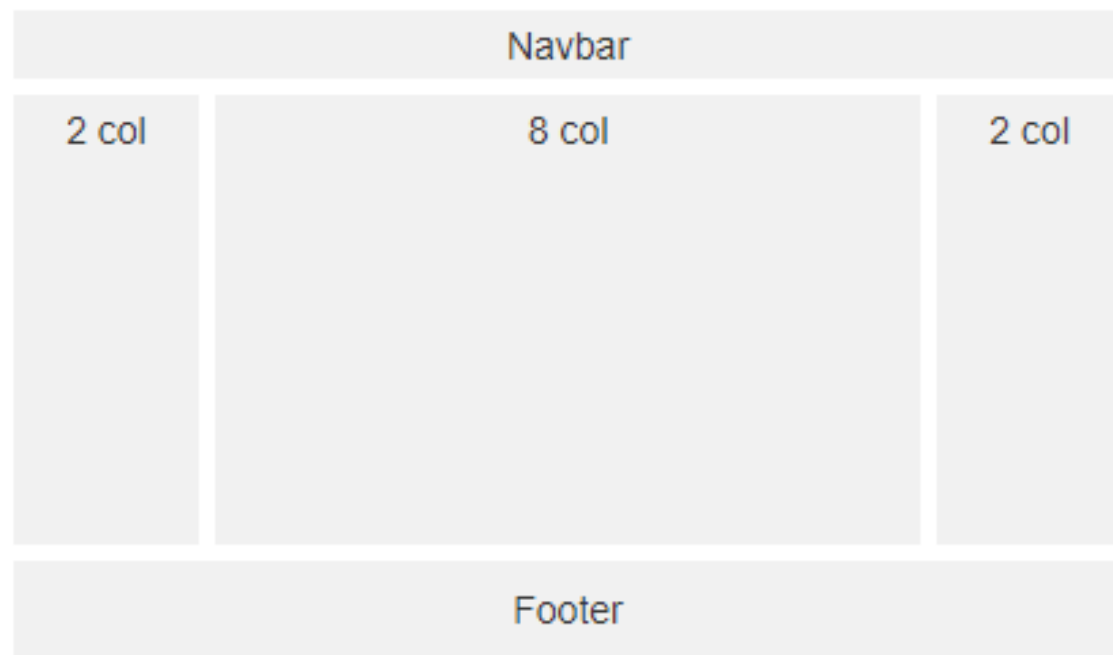
3 col

3 col

Footer

Try it Yourself »

Webpage:



[Try it Yourself »](#)

## Social:

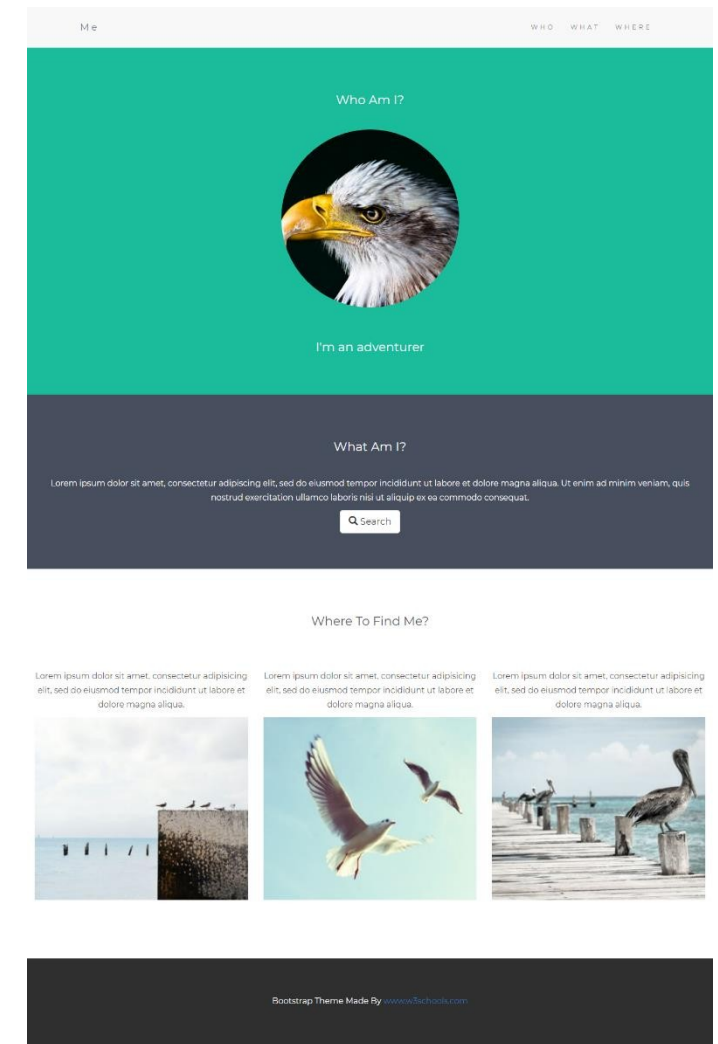
| Navbar |        |       |  |   |
|--------|--------|-------|--|---|
| 3      | 7 - 12 |       |  | 2 |
| 3      | 3      | 7 - 9 |  | 2 |
|        | 3      | 7 - 9 |  |   |
| 3      | 3      | 7 - 9 |  | 2 |
|        | 3      | 7 - 9 |  |   |
| Footer |        |       |  |   |

Try it Yourself »

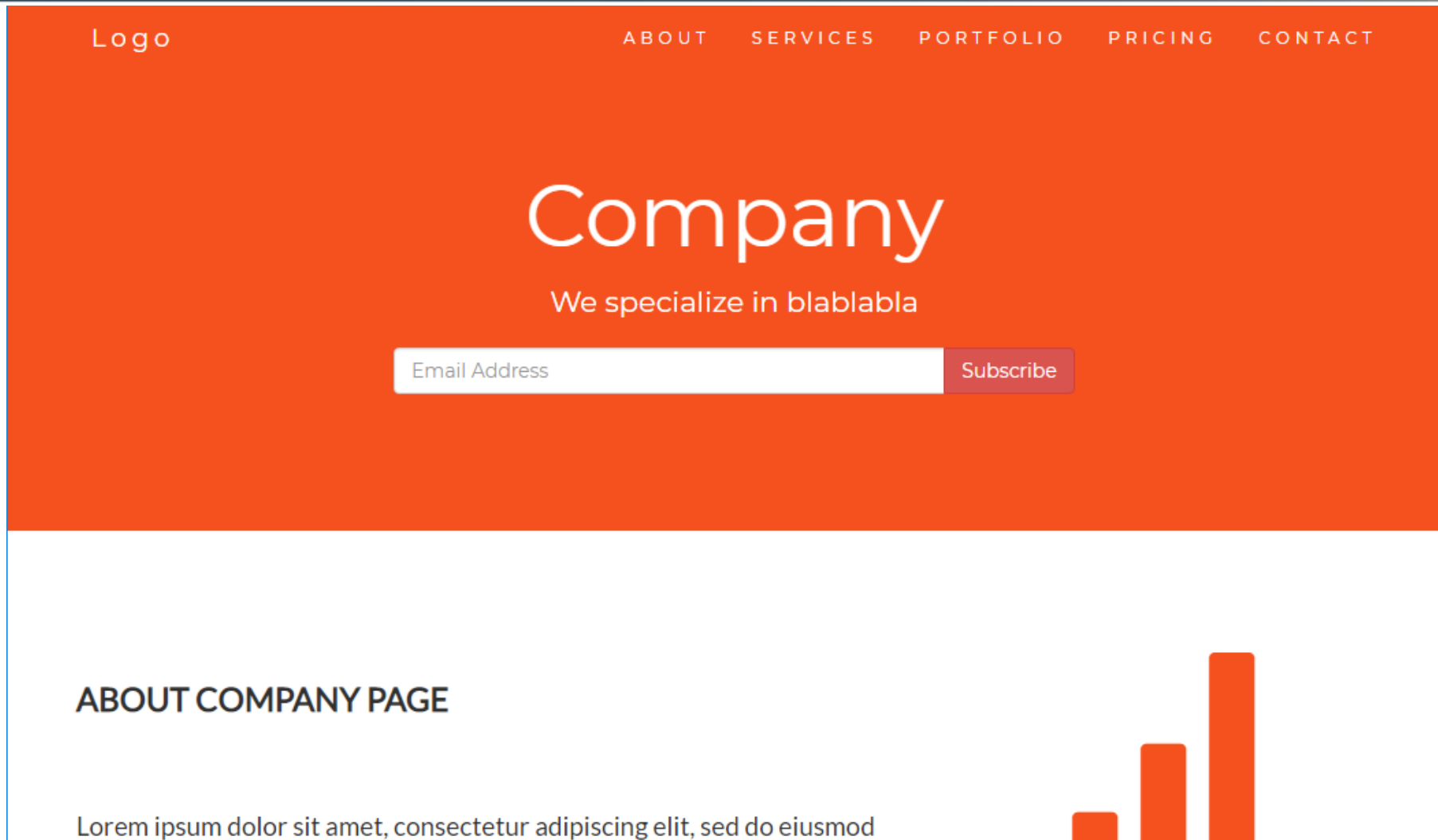
# Bootstrap Theme "Simply Me"



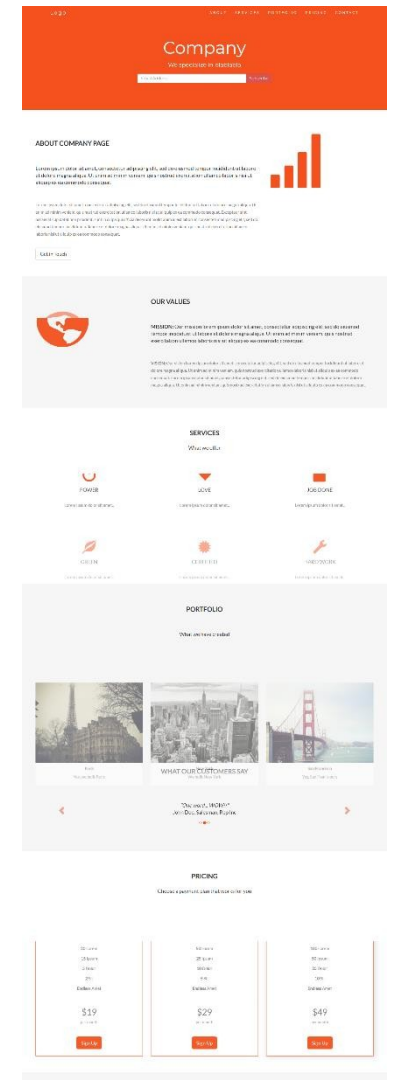
[https://www.w3schools.com/bootstrap/bootstrap\\_theme\\_me.asp](https://www.w3schools.com/bootstrap/bootstrap_theme_me.asp)



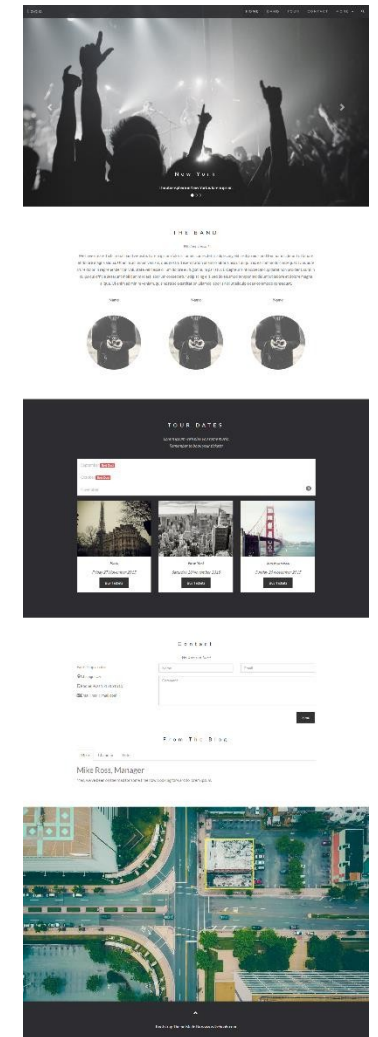
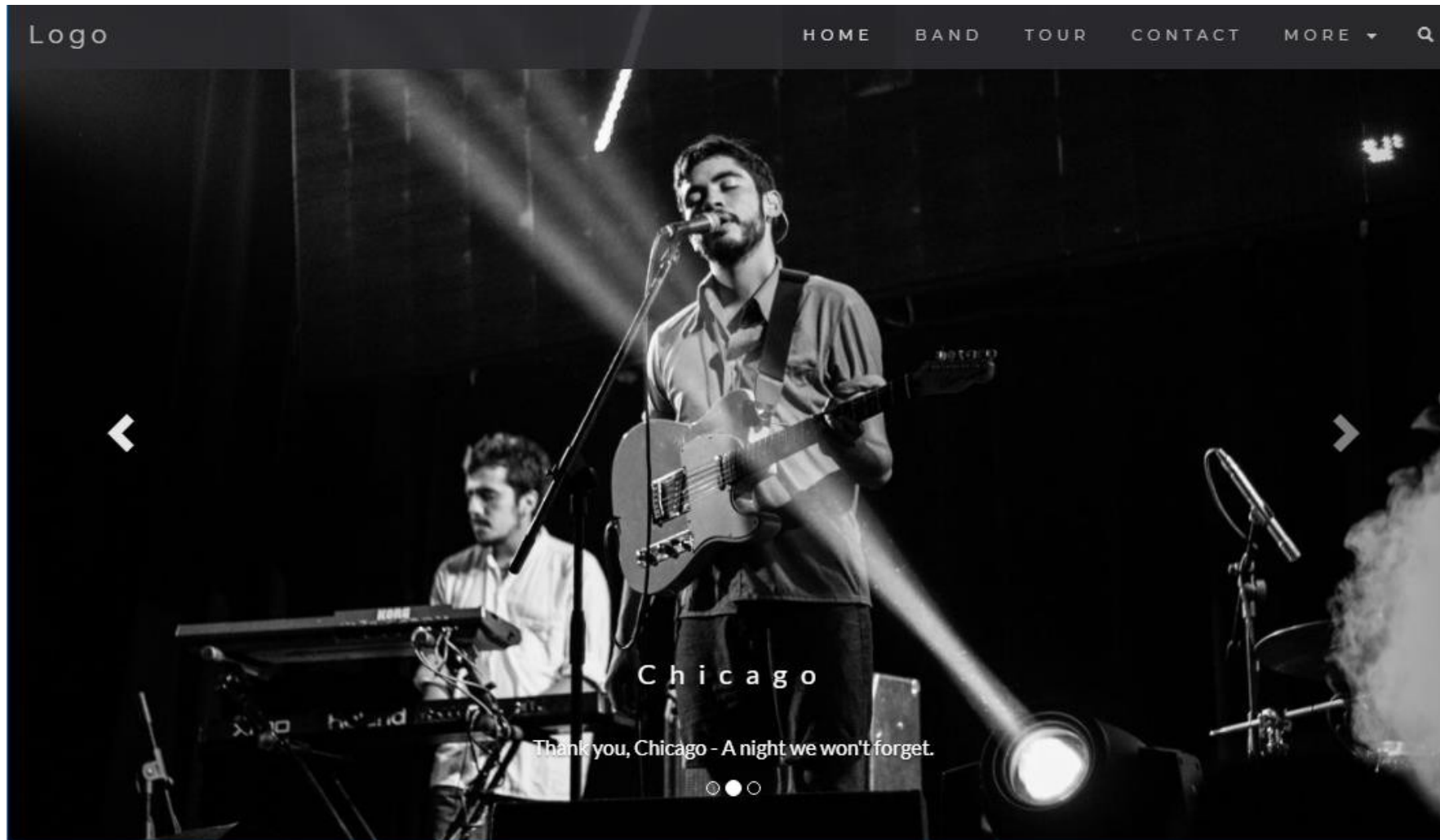
# Bootstrap Theme "Company"



[https://www.w3schools.com/bootstrap/bootstrap\\_theme\\_company.asp](https://www.w3schools.com/bootstrap/bootstrap_theme_company.asp)



# Bootstrap Theme "The Band"



[https://www.w3schools.com/bootstrap/bootstrap\\_theme\\_band.asp](https://www.w3schools.com/bootstrap/bootstrap_theme_band.asp)



# Nhớ gì ?!!!



# Tài liệu tham khảo

- w3schools, “**Bootstrap 3 Tutorial**”,  
<https://www.w3schools.com/bootstrap/default.asp>, 23:20 , 18/08/2018



**Working hard,  
Success in the future !**